Efficiency is ratio of output energy/input energy = mg sinα/F

6 Answer: C

Volume of cylinder = πr2h
Volume of hollowed cylinder = $\pi h(R^2 - r^2)$
Mass of concrete hollowed cylinder = ρV
Therefore change in PE of hollowed concrete cylinder = ρVgΔh
$= (2.0 \times 10^3)(\pi)(10)(0.30^2 - 0.15^2)(9.81)(5.0) = 208 \text{ kJ} \sim 210 \text{ kJ}$

7 Answer: B